

# Neel Gahalot

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## PROFESSIONAL SUMMARY

Innovative and solutions-driven Applied Machine Learning Scientist specializing in AI and Deep Learning, with a proven track record in developing cutting-edge technology in video and audio understanding. Expertise in leveraging large-scale pretraining and multi-modal alignment to achieve state-of-the-art performance. Experienced in deploying scalable models as API services and proficient in leading small, agile teams from project conception through to deployment. Adept at using cloud platforms, CI/CD pipelines, and monitoring solutions to ensure robust performance. Passionate about applying AI to novel and underexplored domains to create transformative solutions. Holds a Master's degree in Computer Engineering with a focus on Machine Learning.

## SKILLS

**Programming Languages:** Python, Bash, SQL

**Machine Learning Framework & Tools:** Pytorch, Jupyter, Airflow, MLflow

**Deep Learning Models:** Whisper, Segment Anything Model, Deeplabv3+, Yolov9, Grounding Dino, TimeSformer, VideoMAE

**CI/CD:** GitHub Actions, GitLab CI

**Cloud Platform:** Google Cloud Platform (GCP), Amazon Web Services (AWS)

**Miscellaneous:** Fine-Tuning, Multi-Modal Training, Large-Scale Pretraining, Transfer Learning, Low Rank Adaptation, Object Detection, Segmentation, Generative AI, BERT, CNN, RNN, LSTM, Transformer

## EXPERIENCE

### Olive Intelligence

New York City

#### Machine Learning Research Scientist

Jun 2024 - Current

Olive Intelligence is an AI Research Company, part of Antler's 2024 residency cohort, building world's first AI Speech Language Therapist.

#### Audio & Language Understanding:

- Leveraged large-scale frequency masking-based pretraining to enhance state-of-the-art performance (arXiv:2311.05203) in Disfluency Classification on the SEP-28K (Apple Stuttering dataset), achieving a 7.2% improvement in F1 scores on the Blocks category with Whisper Tiny's Encoder weights vis-à-vis no pretraining.
- Deployed a Breath Detection Model based on Frame-Wise Breath Detection with Self-Training (arXiv:2402.00288) using Modal as an API service. Used TorchAudio over Librosa with vectorized operations for ZCR (Zero Crossing Rate) extraction to speed up the feature extraction by 63.4%.
- Enhanced phoneme-level alignment in WhisperX by integrating cross-attention weights from Whisper's decoder with Dynamic Time Warping (DTW), improving word-phoneme correspondence accuracy and resolving backtracking failures in forced alignment using Wav2Vec2-ASR-Base-960h.
- Designed a disfluency detection system integrating word-to-phoneme alignment and vocal markers (F0, RMS, Formant) to timestamp utterances; extracted phoneme-level features for processing with a large language model (LLM) to generate insights on speech patterns, tonality, and confidence.

#### Video Understanding:

- Implemented a face-cropping pipeline using Mediapipe and extracted video embeddings with a ResNet backbone; trained a Transformer-based model with positional encoding on these features to classify 3-second video clips (30 FPS) as fluent or disfluent.

#### Multimodal Training:

- Developed a late-fusion multimodal model integrating a Whisper-based audio classifier and a Transformer-based video classifier for disfluency classification; implemented a cross-modal attention mechanism to attend to fused audio-video features before feeding them to a classification layer.

### Greenstand

New York City

#### Deep Learning Engineer, Project Lead - Image Quality Analysis

Aug 2023 – Jun 2024

Greenstand is an environmental services platform that offers tokens to tree planters across the world. Lead a small team in automating the approval of tree captures by implementing a custom Deeplabv3+ solution.

- Engineered an annotation workflow, featuring automated prompting by combining Grounding Dino and Segment Anything Model (SAM), manual refinement via interactive IPywidgets, and COCO conversion with uploads to annotation platforms like Roboflow.
- Implemented Dense Conditional Random Field (CRF) to post-process segmentations from a Deeplabv3+ checkpoint, extracting bounding boxes to generate additional labels with SAM and train a more robust and efficient checkpoint, enhancing inference metrics by 6.24%.
- Trained a DeepLabV3+ model with a Mobilenet backbone, applying Low Rank Adaptation on region-specific datasets, resulting in a 6.49% improvement in foreground Intersection over Union (IOU) compared to a backbone freezing fine-tuning mechanism.

## EDUCATION

### NEW YORK UNIVERSITY

New York City

Master of Science in Computer Engineering

[Relevant Coursework – Machine Learning, Deep Learning, Machine Learning for Cybersecurity, Data Structures and Algorithm]